

Roberto Carlos Barajas

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Career Objectives

Electrical Engineer specializing in power distribution, control hardware, and electrical testing. Proficient in reading complex schematics, component-level circuit troubleshooting, and precise mechanical assembly. Experienced in power electronics, PCB design, and technical documentation.

Education

California State University Fresno, GPA: 2.86 <i>Bachelor's in science in Electrical Engineering</i> - IEEE Fresno State, AIAA Fresno State - Relevant Coursework: Electrical Power (ECE 151), Power Electronics (ECE 153), Programmable Logic Controllers (ECE 119L), Control Systems (ECE 155), Advanced Circuits (ECE 102)	Fresno, CA Aug. 2021 – May 2026
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Experience

Battery Monitoring System with Passive Balancing <i>California State University, Fresno</i> - Designed a 10s3p battery management system using an STM32F030K6Tx microcontroller. - Integrated a MAX14921 analog front end and 14-bit ADC for cell voltage monitoring. - Designed control hardware with CAN bus and RS-485 communication. - Implemented 12V, 5V, and 3.3V regulated power rails from a high-voltage input. - Added optical isolation, solid-state relays, and thermistor-based temperature protection. - Tested and debugged voltage sensing, balancing, and protection functions.	August 2025 – May 2026 Fresno, CA
Avionics Team Member <i>American Institute of Aeronautics and Astronautics Fresno State Chapter</i> - Supported design of electrical power distribution for avionics and recovery bay systems. - Worked with the team on subsystem integration and power routing for flight electronics. - Contributed to electrical design decisions for reliable system operation.	Oct. 2024 – Present Fresno, CA

Certifications

Certificate of Completion <i>PCB design with KiCad - updated for KiCad 8</i> 27.5 Hour KiCad Design Course provided on Udeemy created by Dr Peter Dalmaris	Nov. 2024 Remote
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